Amlogic Hybrid Android S Release Note

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Content

1. Basic Information

1.1. Introduction

This document is the release notes of Amlogic reference source codes aligned with Android S running on Amlogic S905X4/S905C2/S905Y4/S905C3/S805X2 reference hardware.

S905X4/S905C2/S905Y4/S905C3/S805X2 released the GTVS clean version(Hybrid Candidate).

The purpose of this release is:

1. GTVS Clean.
    - (1) CTS(12.0 R4)
    - (2) GTS(9.0 R2)
    - (3) VTS(12.0 R4)
    - (4) CTS-verify(12.0 R4)
    - (5) CTS-on-gsi(12.0 R4)
    - (6) STS(2022-08)
    - (7) TVTS(2.3 R1)
    - (8) Android TV smoking(Youtube testing)
2. NTS pre-cert
3. Pre-certification
    - (1) VMX-IPTV Client E2E
    - (2) VMX-WEB Client E2E
    - (3) VMX-DVB Client E2E(DTVKit)
    - (1) Nagra TKL Simbad for S905X4(DTVKit)
    - (2) Nagra NOCS Simbad for S905C2(DTVKit)
    - (3) Nagra NOCS CAK
    - (4) Irdeto for S905C2/S905C3(Irdeto middware)
    - (5) DDP V4.8 for DVB
    - (6) MS12 V2 for DVB
    - (7) MS12 V1 DVB for S905X4
    - (8) Dolby vision v2.6
    - (9) DTS-HD
    - (10) ART
4. Full function ready:
    - (1) Component compatibility
        - (1) HDMI
        - (2) CVBS
        - (3) WIFI
        - (4) BT
        - (5) Ethernet
        - (6) Bluetooth
        - (7) USB
        - (8) Remote
        - (9) Spdif
    - (2) Suspend/Resume
    - (3) Multi-media functionality.
    - (4) Multi-instance.
    - (5) Widevine CAS.
    - (6) Display functionality.
    - (7) PQ/DI.
    - (8) HDR10.
    - (9) Dolby Vision.
    - (10) System functionality.

(11) Audio-output.
(12) Secure boot V3.
(13) Fastboot
(14) A/B update
(15) AVB Anti-roll back
(16) HDMI CTS 2.0
(17) HDMI CEC
(18) HDR10+
(19) HDCP 1.4/2.3

5. Performance
   (1) Benchmark
   (2) UX
   (3) APP launch speed
   (4) Video start/Switch speed
   (5) Memory usage
6. DTVKit
   (1) DVB-C Program search
   (2) DVB-S/DVB-S2 Program search
   (3) DVB-T/DVB-T2 Program search
   (4) Channel auto search
   (5) Channel manual search
   (6) NIT search
   (7) MHEG5
   (8) Teletext
   (9) PVR
   (10) TimeShift
   (11) subtitle
   (12) Program Edit
   (13) LCN
   (14) EPG
   (15) Channels Lock
   (16) Parental Lock
   (17) AD
   (18) Info Banner
   (19) Aspect Ratio
   (20) update channel
   (21) Date、Time
   (22) Audio language
   (23) Switch Channel
   (24) FCC/PIP

The known issues of this release is:
1. TrueHD has some compatibility issues.
   Quick and repeated seek has exit or struck.
2. MS12 has some compatibility issues.
3. DVB has some stability issues in VMX FCC/PIP cases.
4. GooglePermissionController APK caused some xTS cases failed and ExoPlayer open failed.

1.2. Chip Information

S905X4/S905C2/S905Y4/S905C3/S805X2 :

We choose S905X4/S905Y4 reference platform as our major developing and testing platforms for this release.

Item	**S905X4/S905C2**	**S805X2**
CPU	Quad-core ARM Cortex-A55	Quad-core ARM Cortex-A35
GPU	ARM Mali-G31 MP2	ARM Mali-G31 MP2
Memory	32-bit DDR3/4/3L, LPDDR3/4	32-bit DDR3/4/3L, LPDDR4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps	AV1/H.265/VP9/AVS2-P2 to 1080p@60fps
Video Encoding	H.265/JPEG to 1080p@60fps	SW encoder
HDMI-TX	4k60hz	1080p60hz
AV output	CVBS	CVBS
Ethernet	10M/100M/1000M	10M/100M
IP License	Dolby Audio/DTS/Dolby Vision	Dolby Audio/DTS
Demux	Support	Support

Item	**S905Y4/S905C3**
CPU	Quad-core ARM Cortex-A35
GPU	ARM Mali-G31 MP2
Memory	32-bit DDR3/4/3L, LPDDR4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps
Video Encoding	SW encoder
HDMI-TX	4k60hz
AV output	CVBS
Ethernet	10M/100M
IP License	Dolby Audio/DTS/Dolby Vision
Demux	Support

1.3. Reference Platform

Mbox:

- **Ohm(S905X4)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Ohmcas(S905C2)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Oppen(S905Y4)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Oppencas(S905C3)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Planck(S805X2)**
  EMMC,WIFI QCA6174 ,DDR4 1GB
- **Planck_gtv(S805X2)**
  EMMC,WIFI QCA6174 ,DDR4 1.5GB

1.4. How to Get Code

You can download Android S source code by running the following repo commands:

ODM or OEM

Note: Permission for downloading code

1. Google server(Whole code that can build the final firmware, Kernel/uboot are the binary)
2. Amlogic Openlinux server(Kernel/uboot source code)

1. use below command to get the code in google server(apply the license from Google):

```
$cd~/<your-google-repo-dir>/
$repo init -u
https://partner-android.googlesource.com/platform/vendor/pdk/ohm/ohm-userdebug/manifest
-b builds/s-tv-amlogic-hybrid-release/STTC.220803.001
$repo sync
```

2. use below command to get the code in amlogic server(apply the license from Amlogic)

```
$cd~/<your-amlogic-repo-dir>/
$repo init -u ssh://git@openlinux.amlogic.com/s-amlogic/platform/manifest.git -b s-amlogic -m
openlinux/ott/amlogic/s-amlogic-source-20220805-gtvs.xml
$repo sync
```

Notice: Oversea customer use below command:

```
$cd~/<your-amlogic-repo-dir>/
$repo init -u ssh://git@openlinux2.amlogic.com/s-amlogic/platform/manifest.git -b s-amlogic
-m openlinux/ott/amlogic/s-amlogic-source-20220805-gtvs.xml
$repo sync
```

Needed storage space: need about 200G storage space.

1.5. How to integrate the code

Some customers need to build the UBoot/kernel source code . Use the below commands to integrate the code:

```
$cd ~/<your-amlogic-repo-dir>/
$cp -rf bootloader                        <your-google-repo-dir>/
cp -rf build                                        <your-google-repo-dir>/
cp -rf common/                                      <your-google-repo-dir>/
cp -rf device/amlogic/ohmcas         <your-google-repo-dir>/device/amlogic/
cp -rf device/amlogic/ohmcas-kernel         <your-google-repo-dir>/device/amlogic/
cp -rf device/amlogic/oppencas         <your-google-repo-dir>/device/amlogic/
cp -rf device/amlogic/oppencas-kernel         <your-google-repo-dir>/device/amlogic/
cp -rf vendor/wifi_driver                                 <your-google-repo-dir>/vendor/
cp -rf hardware/amlogic/media_modules/            <your-google-repo-dir>/hardware/amlogic/
cp -rf vendor/amlogic/common/gpu/                      <your-google-repo-dir>/vendor/amlogic/common/
cp -rf vendor/amlogic/common/apps/Updater       <your-google-repo-dir>/vendor/amlogic/common/apps/
cp -rf vendor/amlogic/common/tdk_linuxdriver/       <your-google-repo-dir>/vendor/amlogic/common/
cp -rf vendor/amlogic/common/pre_submit_for_google      <your-google-repo-dir>/vendor/amlogic/common/
cp -rf vendor/amlogic/common/patches      <your-google-repo-dir>/vendor/amlogic/common/
cp -rf vendor/amlogic/common/external/exfat             <your-google-repo-dir>/vendor/amlogic/common/external/
cp -rf vendor/amlogic/common/external/ntfs-3g            <your-google-repo-dir>/vendor/amlogic/common/external/
cp -rf vendor/amlogic/ohmcas/                      <your-google-repo-dir>/vendor/amlogic/
cp -rf vendor/amlogic/oppencas/                      <your-google-repo-dir>/vendor/amlogic/
rm -rf <your-google-repo-dir>/vendor/amlogic/common/external/ffmpeg
cp -rf vendor/amlogic/common/external/ffmpeg   <your-google-repo-dir>/vendor/amlogic/common/external/

cd <your-google-repo-dir>/build/make/; git checkout -f
```

1.6. Dynamic fingerprint

Although Android TV devices are similar, their system properties can differ. To help reduce the build load, the Android TV OEM partition has been released for use with Android TV on Android 12.

```
1.  Change device/amlogic/oppen/oem/oem.prop
. build/envsetup.sh
lunch oppen-user
make custom_images -j8

2.  copy out/target/product/oppen/oem.img to device/amlogic/oppen/oem/

As ohm.
```

1.7. How to config DDP MS12 DTS Dolby Vision and HDCP

Dolby Digital Plus, Dolby ms12, DTS or Dolby vision is stored in the oem partition.

1. Copy libHwAudio_dcvdec.so to vendor/amlogic/common/prebuilt/libstagefrighthw/lib/libHwAudio_dcvdec.so
2. Copy libdolbyms12.so to device/amlogic/common/dolby_ms12/install/encrypted_lib/libdolbyms12.so
3. Copy libdolbyms12.so(ms12v1) to device/amlogic/common/dolby_ms12/install/encrypted_lib/ms12v1/libdolbyms12.so
4. Copy libHwAudio_dtshd.so to vendor/amlogic/common/prebuilt/libstagefrighthw/lib/libHwAudio_dtshd.so
5. Copy dovi.ko to device/amlogic/ohm/files/dovi.ko , as ohmcas/oppen etc.
6. Copy dovi_fw.bin to device/amlogic/ohm/dovi_fw.bin , as ohmcas/oppen etc.
7. Copy hdcp_tx22 and firmware.le to vendor/amlogic/restricted_libs/hdcp_tx22/

Enable Dolby Digital Plus, Dolby ms12, DTS or Dolby vision Macro.

```
TARGET_BUILD_WITH_DOVI :=true
TARGET_DOLBY_VERSION := ms12_v2      or set ms12_v1 ddp_only
TARGET_DTS_VERSION :=dtshd
```

Generate OEM partition

```
. build/envsetup.sh
lunch ohm_hybrid-user
make custom_images -j8

copy out/target/product/ohm/oem.img to device/amlogic/ohm/oem/oem_hybrid/

As oppen etc.
```

1.8. How to Build Code

Before build the code, you should execute this command at the code root directory for merge some important patches.

./vendor/amlogic/common/patches/detect.sh

Build Bootloader

1. ohm_hybrid(S905X4)

```
cd bootloader/uboot-repo/
./mk sc2_ah212 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/ohm/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohm/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohm/upgrade/
```

2. ohmcas(S905C2)

```
cd bootloader/uboot-repo/
./mk sc2_ah232 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/ohmcas/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohmcas/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohmcas/upgrade/
```

3. oppen_hybrid(S905Y4)

```
cd bootloader/uboot-repo/
./mk s4_ap222 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/oppen/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/oppen/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/oppen/upgrade/
```

4. oppencas(S905C3)

```
cd bootloader/uboot-repo/
./mk s4_ap232 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/oppencas/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/oppencas/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/oppencas/upgrade/
```

5. planck/planck_gtv(S805X2)

```
cd bootloader/uboot-repo/
./mk s4_aq222 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/planck/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/planck/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/planck/upgrade/
```

Noticed: Bootloader disabled fastboot unlock function by default. Android VTS/GSI or debug need enable fastboot unlock function through --fastboot-write. Like ./mk s4_ap222 --avb2 --vab --fastboot-write.

Build Kernel

1. ohm_hybrid(S905X4)

```
./device/amlogic/common/kernelbuild/mk ohm -v 5.4 -t user
```

2. ohmcas(S905C2)

```
./device/amlogic/common/kernelbuild/mk ohmcas -v 5.4 -t user
```

3. oppen_hybrid(S905Y4)

```
./device/amlogic/common/kernelbuild/mk oppen -v 5.4 -t user
```

4. oppencas(S905C3)

```
./device/amlogic/common/kernelbuild/mk oppencas -v 5.4 -t user
```

5. planck_gtv(S805X2)

```
./device/amlogic/common/kernelbuild/mk planck -v 5.4 -t user
```

6. planck(S805X2)

```
export KERNEL_A32_SUPPORT=true
./device/amlogic/common/kernelbuild/mk planck -v 5.4 -t user
```

Build Android

1. ohm_hybrid(S905X4)

```
. build/envsetup.sh
lunch ohm_hybrid-user
make -j8
```

2. ohmcas_gtv(S905C2)

```
. build/envsetup.sh
lunch ohmcas_gtv-user
make -j8
```

3. oppen_hybrid(S905Y4)

```
. build/envsetup.sh
lunch oppen_hybrid-user
make -j8
```

4. oppencas_gtv(S905C3)

```
. build/envsetup.sh
lunch oppencas_gtv-user
make -j8
```

5. planck_gtv(S805X2)

```
. build/envsetup.sh
lunch planck_gtv-user
make -j8
```

6. planck(S805X2)

```
. build/envsetup.sh
export KERNEL_A32_SUPPORT=true
lunch planck-user
make -j8
```

Noticed: Android S will start to use kernel 5.4 and we setup a new kernel and modules build way named standalone build.

Standalone build is a standard kernel and modules build way. We need separate Android and Kernel build.

Android S has two UI: Waston UI(Android TV) and Amati UI(Google TV).

Lunch project name	Type
ohm	OTT ATV
ohm_gtv	OTT GTV
ohm_hybrid	DVB GTV
ohmcas	DVB ATV
ohmcas_gtv	DVB GTV
oppen	OTT ATV
oppen_gtv	OTT GTV
oppen_hybrid	DVB GTV
oppencas	DVB ATV
oppencas_gtv	DVB GTV
planck	OTT ATV
planck_gtv	OTT GTV

1.9. Device Vendor SCS(Secure boot v3)

Device Vendor SCS(Secure boot v3), please refer the documents.

- <Amlogic S905X4 Series Secure Boot User Guide>
- <Amlogic S905Y4 Series Secure Boot User Guide>
- <Amlogic Linux SCS Signing ToolUserGuide>

1.10. Secure Mode

This document provides required security mode configuration guidelines to prevent system data from being tampering with. Please refer the documents:
<Security Mode Configuration Guide>

1.11. Fastboot

We MUST update adb/fastboot version to 29.0.1-5644136 or 29.0.1-5644136+. Otherwise there are lots of adb/flash issues.

adb:

```
Android Debug Bridge version 1.0.41
Version 29.0.1-5644136
```

fastboot

```
fastboot version 29.0.1-5644136
```

You can download latest SDK Platform-Tools from
https://developer.android.com/studio/releases/platform-tools.html#downloads

1.12. *How to Upgrade*

There are 3 ways for update.

- Upgrade with USB burn tool
- Upgrade with OTA
- Upgrade with fastboot

1.12.1. all partitions

If you like to use our usb burn tool to update the images, it's the same as before.We have a fastboot flash-all script to update image, just like Android S. You need not care about the dynamic partitions.

Partition name	make command	imgae name	burning ways
all partitions	make otapackage	1.ohm-ota-xxxxxxxx.zip 2.aml_upgrade_package.img 3.ohm-fastboot-xxxxxxxx.zip	1.by OTA upgrade 2.by USB burning 3.by fastboot
	make	aml_upgrade_package.img	by USB burning

1.12.2. physical partitions

In Android S, physical partitions is bootloader, boot, logo, vendor_boot etc.
Fastboot should be in bootloader mode when flash these physical partitions.
Below is the way of burning each partitions:

Partition name	make command	image name	burning ways
boot	make bootimage	boot.img	fastboot flash boot boot.img
logo	make logoimg	logo.img	fastboot flash logo logo.img
uboot	./mk sc2_ah212 --avb2 --vab	u-boot.bin.signed	fastboot flash bootloader u-boot.bin.signed
odm_ext	make odm_ext_image	odm_ext.img	fastboot flash odm_ext odm_ext.img
vendor_boot	make vendorbootimage	vendor_boot.img	fastboot flash vendor_boot vendor_boot.img

Notes: execute ‘fastboot flashing unlock’ to unlock the device if wants to use fastboot commands.

1.12.3. logical partitions

In Android S, logical partitions is system, system_ext,vendor, odm, product.
Fastboot should be in fastbootd mode when flash these logical partitions.
Below is the way of burning each partitions:

Partition name	make command	image name	burning ways
system	make systemimage	system.img	fastboot flash system system.img
vendor	make vendorimage	vendor.img	fastboot flash vendor vendor.img
odm	make odmimage	odm.img	fastboot flash odm odm.img
product	make productimage	product.img	fastboot flash product product.img
system_ext	make systemextimage	system_ext.img	fastboot flash system_ext system_ext.img

Notes: execute ‘fastboot flashing unlock’ to unlock the device if wants to use fastboot commands.

2. Test Report

2.1. Auto Reboot in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AH212#175	Auto reboot	72H	Pass
AH212#186	Auto reboot	72H	Pass
AH212#197	Auto reboot	72H	Pass
AH212#237	Auto reboot	72H	Pass
AP222#200	Auto reboot	72H	Pass
AP222#224	Auto reboot	72H	Pass
AP222#228	Auto reboot	72H	Pass
AP222#231	Auto reboot	72H	Pass
AH212#200	uboot autoreboot	72H	Pass
AH212#238	uboot autoreboot	72H	Pass
AH212#248	uboot autoreboot	72H	Pass
AH212#249	uboot autoreboot	72H	Pass
AP222#146	uboot autoreboot	72H	Pass
AP222#164	uboot autoreboot	72H	Pass
AP222#313	uboot autoreboot	72H	Pass
AP222#319	uboot autoreboot	72H	Pass

2.2. Auto Shutdown/Suspend in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AH212#200	Auto shutdown	72H	Pass
AH212#238	Auto shutdown	72H	Pass
AH212#248	Auto shutdown	72H	Pass
AH212#249	Auto shutdown	72H	Pass
AP222#200	Auto shutdown	72H	Pass
AP222#224	Auto shutdown	72H	Pass
AP222#228	Auto shutdown	72H	Pass
AP222#231	Auto shutdown	72H	Pass
AH212#175	Auto suspend	72H	Pass
AH212#186	Auto suspend	72H	Pass
AH212#197	Auto suspend	72H	Pass
AH212#237	Auto suspend	72H	Pass
AP222#200	Auto suspend	72H	Pass
AP222#224	Auto suspend	72H	Pass
AP222#228	Auto suspend	72H	Pass
AP222#231	Auto suspend	72H	Pass

2.3. 3D Game burning Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#175	3D Game	72H	Pass
AH212#186	3D Game	72H	Pass
AH212#197	3D Game	72H	Pass
AH212#237	3D Game	72H	Pass

AP222#200	3D Game	72H	Pass
AP222#224	3D Game	72H	Pass
AP222#228	3D Game	72H	Pass
AP222#231	3D Game	72H	Pass

2.4. Monkey test Test Report

Test Platform	Test Case	Total test time	Test Result	Comments
AH212#200	Monkey test/300ms	72H	Pass	
AH212#238	Monkey test/300ms	72H	Pass	
AH212#248	Monkey test/300ms	72H	Pass	
AH212#249	Monkey test/300ms	72H	Pass	
AP222#146	Monkey test/300ms	72H	Pass	
AP222#164	Monkey test/300ms	72H	Pass	
AP222#313	Monkey test/300ms	72H	Pass	
AP222#319	Monkey test/300ms	72H	Pass	

Monkey Black list
com.droidlogic.PPPoEcom.droidlogic.PPPoE
com.droidlogic.miracast
com.droidlogic.mediacenter
com.droidlogic.otaupgrade
com.droidlogic.tvinput
com.hpplay.happyplay.aw
com.android.tv.settings
com.droidlogic.tv.settings
com.google.android.tungsten.setupwraith
com.google.android.gms
com.android.vending
com.google.android.youtube.tv
com.google.android.videos
com.google.android.tv.remote.service
com.google.android.tvrecommendations
com.google.android.gms.persistent
com.google.android.youtube.tv.recommendations
com.google.android.tv.frameworkpackagestubs
com.google.android.tvcom.droidlogic.videoplayer

2.5. FactoryReset Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#200	Factory reset	72H	Pass
AH212#238	Factory reset	72H	Pass
AH212#248	Factory reset	72H	Pass
AH212#249	Factory reset	72H	Pass
AP222#200	Factory reset	72H	Pass
AP222#224	Factory reset	72H	Pass
AP222#228	Factory reset	72H	Pass
AP222#231	Factory reset	72H	Pass

2.6. A/V Burn Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#94	4K H265	72H	Pass
AH212#128	4K H265	72H	Pass
AH212#137	4K H265	72H	Pass
AH212#194	4K H265	72H	Pass
AP232#134	4K H265	72H	Pass
AH212#128	4K H264	72H	Pass
AH212#238	4K H264	72H	Pass
AP232#131	4K H264	72H	Pass
AH212#194	4K VP9	72H	Pass
AH212#249	4K VP9	72H	Pass
AP232#135	4K VP9	72H	Pass
AP232#130	4K AV1	72H	Pass
AH212#105	4K AV1	72H	Pass
AP222#176	4K AV1	72H	Pass
AP222#134	4K H265	72H	Pass
AP222#169	4K H264	72H	Pass
AP222#200	4K VP9	72H	Pass
AH212#120	Youtube	12H	Pass
AH212#194	Youtube	12H	Pass
AH212#120	Youtube	12H	Pass
AP222#99	Youtube	12H	Pass
AP222#188	full format	72H	Pass
AP222#159	full format	72H	Pass
AP222#169	full format	72H	Pass

2.7. DVB Stress Test

Test Platform	Test Case	Total test time	Test Result	Comments
AH212#637	Auto Search	72H	Pass	
AH212#640	Auto Search	72H	Pass	
AH212#126	Auto Search	48H	Fail	DVB/C automatic search page responds to TV sound
AH212#169	Auto Search	72H	Pass	
AH212#166	4K Bitstream Longterm Playing	72H	Pass	
AH212#232	4K Bitstream Longterm Playing	72H	Pass	
AH212#126	4K Bitstream Longterm Playing	72H	Pass	
AH212#169	4K Bitstream Longterm Playing	72H	Pass	
AH212#166	Single freq. Channel Change	72H	Pass	
AH212#232	Single freq. Channel Change	72H	Pass	
AH212#126	Single freq. Channel Change	72H	Pass	
AH212#169	Single freq. Channel	72H	Pass	

	Change			
AH212#169	Multi freq. Channel Change	72H	Pass	
AH212#166	Multi freq. Channel Change	12H	Fail	Exit live tv
AH212#232	Multi freq. Channel Change	72H	Pass	
AH212#126	Multi freq. Channel Change	72H	Pass	
AH212#169	Subtitle Longterm Playing	72H	Pass	
AH212#166	Subtitle Longterm Playing	72H	Pass	
AH212#232	Subtitle Longterm Playing	72H	Pass	
AH212#126	Subtitle Longterm Playing	72H	Pass	
AH212#1	Program Switch(FCC)	20H	**Fail**	TV black screen with sound
AH212#2	Program Switch(FCC)	20H	**Fail**	TV black screen with sound
AH212#1	LongTime Playback(PIP)	12H	**Fail**	subtitles are not automatically refreshed
AH212#2	LongTime Playback(PIP)	72H	Pass	
AH212#1	Program Switch(PIP)	72H	Pass	
AH212#2	Program Switch(PIP)	72H	Pass	

2.8. Video Format Test Report

Noticed: Please confirm the license before opening the multimedia format.

Extension	Codec Detail	Tested Resolution	Ohm/Ohmcas	Oppen/Oppencas	Planck
.3g2	H263	704x576	Support	Support	Support
.3gp	MPEG-4 Visual	640x480	Support	Support	Support
	MPEG-4	320x240	Support	Support	Support
	H263	320x276	Support	Support	Support
	H264	1920x1080	Support	Support	Support
.avi	AVC	1920x1080	Support	Support	Support
	M-JPEG	1024x576	Support	Support	Support
	RealMagic MPEG-4	720x480	Support	Support	Support
		720x576	Support	Support	Support
	h264	1920x1080	Support	Support	Support
	FF mpeg MPEG4	640x480	Support	Support	Support
.dat	MPEG-1	352x288	Support	Support	Support
.evo	VC-1	1920x1080	Support	Support	Support
.f4v	AVC(H264)	1280x720	Support	Support	Support
.flv	Sorenson Spark	320x276	Support	Support	Support
	VP6	800x342	Support	Support	Support
.mp4	AVC(H264)	1920x1080	Support	Support	Support

	HEVC(H265)	1920x1080	Support	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support	No support
	8K HEVC(8K H265)	8192x4320	Support	Support	No support
	AV1	1920x1080	Support	Support	Support
	4K AV1	3840x2160	Support	Support	No support
	H263	176x144	Support	Support	Support
.m2ts	AVC	1920x1080	Support	Support	Support
	VC-1	1920x1080	Support	Support	Support
.m2v	MPEG-2	480x576	Support	Support	Support
.m4v	AVC	1280x720	Support	Support	Support
.mkv	WMV3	1280x720	Support	Support	Support
	MPEG-4 Visual	1920x1080	Support	Support	Support
	AVC	1920x1080	Support	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support	No support
	8K HEVC(8K H265)	8192x4320	Support	Support	No support
	VP8	1920x1080	Support	Support	Support
	VP9	1920x1080	Support	Support	Support
.mov	MPEG-4 Visual	1280x720	Support	Support	Support
	mjpa	640x480	Support	Support	Support
	H263	320x240	Support	Support	Support
	M-JPEG	640x480	Support	Support	Support
	AVC(H264)	1920x1080	Support	Support	Support
	4K H264	3840x2160	Support	Support	No support
.mpeg	MPEG-2	1920x1080	Support	Support	Support
	VC-1	1920x1080	Support	Support	Support
	MPEG-1	720x576	Support	Support	Support
.mts	AVC	1440x1080	Support	Support	Support
.ogm	XVID	640x480	Support	Support	Support
.PMP	H264	480x272	Support	Support	Support
.tp	MPEG-2	1920x1088	Support	Support	Support
.ts	HEVC(H265)	1920x1080	Support	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support	No support
	8K HEVC(8K H265)	8192x4320	Support	Support	No support
	MPEG-1	1920x1080	Support	Support	Support
	avs	720x576	Support	Support	Support
	avs+	1920x1080	Support	Support	Support
	avs2	3840x2160	Support	Support	No support
	AVC	1920x1080	Support	Support	Support
	MPEG-2	1920x1080	Support	Support	Support
.vob	MPEG-2	720x576	Support	Support	Support
.wmv	WMV1	640x480	Support	Support	Support
	WMV2	768x432	Support	Support	Support
	WMV3	1920x1080	Support	Support	Support
.webm	VP8	1920x1080	Support	Support	Support
	VP9	1920x1080	Support	Support	Support
	4K VP9	3840x2160	Support	Support	No support

	8K VP9	8192x4320	Support	Support	No support
	AV1	1920x1080	Support	Support	Support
4K	AVC(H264)	3840x2160	Support	Support	No support
	HEVC(H265)	3840x2160	Support	Support	No support
	VP9	3840x2160	Support	Support	No support
	AV1	3840x2160	Support	Support	No support
8K	HEVC(H265)	8192x4320	Support	Support	No support
	VP9	8192x4320	Support	Support	No support
	AV1	7680x4320	Support	Support	No support

3. Tools version

Tool name	version
USB burning tool	**Aml_Burn_Tool_V3.2.2**
Customization	Amlogic customizationl V.2.1.3
Linux Secure boot V3	Aml_Linux_SCS_SignTool V1.9.0
adb	1.0.41
fastboot	29.0.1
Kernel tool chain	gcc-linaro-6.3.1-2017.02-x86_64_arm-linux-gnueabihf gcc-linaro-6.3.1-2017.02-x86_64_aarch64-linux-gnu

4. License version

Licensed Feature Name	Version	Document version
Dolby Digital Plus for Android	v20220126	In firmware Readme text
Dolby Digital for Android	v20200723	In firmware Readme text
Dolby MS12 V1.3.2 for android	v20220624	In firmware Readme text
Dolby MS12 V2.4 for android	v20220428	Amlogic_Dolby_MS12_v2_release_note_V0.4
DTS-HD for android	v20220614	In firmware Readme text
DRA Decoder for android	v20210728	In firmware Readme text
Secure OS TDK	TDK is included in the code	TDK Integration User Guide (3.3) S905X4 Secure Key Provision User Guide (3.3) S905Y4 Secure Key Provision User Guide (3.3)
Widevine	L1/L3	Amlogic WideVine Integration User Guide v0.9
Playready		Amlogic PlayReady Integrate User Guide v1.32
Dolby Vision 2.6 for Android kernel5.4.180 64bit	Ohm kernel5.4.180 64bit dovi.ko stb_release_20220607 dovi_fw_u2019.bin release-20210413 Oppen kernel5.4.180 64bit dovi.ko stb_release_20220607 dovi_fw_u2019.bin release-20210413	Amlogic DolbyVision Release (v0.5)
HDCP 1.4 for HDMI TX for Android	v20180408	Amlogic_HDCP1.4_Tx_Release_Note_V0.1
HDCP 2.X for HDMI TX for Android	v20201212	Amlogic_HDCP2.2_Tx_Release_Note_V1.0
Verimatrix	vendor/vmx/bootloader(m9x4/openlinux-android-s: 5eefa32c2604da5f9b6d8fb5539594a616b730d5) vendor/vmx/bootloader(m9y4/openlinux-android-s: 69f8abf83bfe7915529fb67c9c8c16a188aacb14) vendor/amlogic/prebuild/verimatrix(r-amlogic: 22f332d22319f62d716c73ef16adb20cf7efcf47) vmx/sdk-rel(m9x4-rel: be555e234f5e86763b55e34aaf00a96cf95da757) vmx/sdk-rel(m9y4-rel: ea3f467a4f2c2d82b3449f6242e134618e1ad8f7) vendor/vmx/vmx-rel(master: d330907e3518121261364f1b8d8c62b9f68b3444)	OHM S905X4 Verimatrix Android SDK Integration User Guide (2.7) S905X4 Verimatrix Android Client Integration User Guide (5.2) Amlogic CAS - Verimatrix Adaption API Guide-v1.0 Oppen S905Y4 Verimatrix Android Client Integration User Guide (7.2) S905Y4 Verimatrix Android SDK Integration User Guide (1.7) Amlogic CAS - Verimatrix Adaption API Guide-v1.0 Planck S805X2 Verimatrix Android Client Integration User Guide (8.2)
Nagra TKL/NOCS	nagra-sdk(projects/openlinux/v2.2: b06540133ed713b3c94c53fbdbf23b5	OHM S905X4 Nagra TKL Android MediaCasV2

	12394968a) nagra-cashal-rel(projects/openlinux/v2.0-android-r:de963e1f2d896f39bdac68433e6eada026a0778c) nagra-sdk(projects/openlinux/v3.4:e2536e72cae003fdbd05b29fe4f13392a52e4dba) nagra-cashal-rel(projects/openlinux/v3.2:211c7484d6d26f3e8a11a244ab8af51e0f3b6153)	Integration User Guide (1.1) S905X4 Nagra TKL Android SDK Integration User Guide (1.2) S905C2 Nagra NOCS Android MediaCasV2 Integration User Guide (1.1) S905C2 Nagra NOCS Android SDK Integration User Guide (1.1)

5. License path

Licensed Files	Path
HDCP TX firmware.le	/odm/etc/firmware/firmware.le
Dolby	/oem/lib/libHwAudio_dcvdec.so
MS12 V2	/oem/lib/ms12/libdolbyms12.so
MS12 V1	/oem/lib/libdolbyms12.so
Dolby vision	/oem/overlay/dovi.ko /oem/firmware/dovi_fw.bin